

```
for (let i = 1, i <= 3, i++) {
```

```
  log("Hello")
```

```
}
```

initialization

condition

log('

update

$i = 1$

$i <= 3$

$\log("Hello")$

$i++$

$i = 1$

$1 <= 3$ True

Hello

$i++ = 1+1 = 2$

$i = 2$

$2 <= 3$ True

Hello

$i = 2+1 = 3$

$i = 3$

$3 <= 3$ True

Hello

$i = 3+1 = 4$

$i = 4$

$4 <= 3$ False

X

X

→ even ~~odd~~ number in range (1, 10)

```
for (let i = 1, i <= 10, i++) {
```

```
  if (i % 2 === 0) {
```

```
    print
```

```
    console.log(i)
```

```
  }
```

initialization

condition

logic

update

$i = 1$

$i \leq 10$

if $i \% 2 == 0$

$i = i + 1$

$i = 1$

$1 \leq 10$ (True)

if $1 \% 2 == 0$ (False)
~~print(1)~~

$i = 1 + 1 = 2$

$i = 2$

$2 \leq 10$ (True)

if $2 \% 2 == 0$ (T)
print(2)

$i = 2 + 1 = 3$

$i = 3$

$3 \leq 10$ (True)

if $3 \% 2 == 0$ (F)

$i = 3 + 1 = 4$

$i = 4$

$4 \leq 10$ (True)

if $4 \% 2 == 0$ (T)

$i = 4 + 1 = 5$

$i = 5$

$5 \leq 10$ (True)

print(4)

if $5 \% 2 == 0$ (F)

$i = 5 + 1 = 6$

$i = 6$

$6 \leq 10$ (True)

if $6 \% 2 == 0$ (T)

print(6)

$i = 6 + 1 = 7$

$i = 7$

$7 \leq 10$ (True)

if $7 \% 2 == 0$ (F)

$i = 7 + 1 = 8$

$i = 8$

$8 \leq 10$ (True)

if $8 \% 2 == 0$ (T)

print(8)

$i = 8 + 1 = 9$

$i = 9$

$9 \leq 10$ (True)

if $9 \% 2 == 0$ (F)

$i = 9 + 1 = 10$

$i = 10$

$10 \leq 10$ (True)

if $10 \% 2 == 0$ (T)

print(10)

$i = 10 + 1 = 11$

$i = 10$

$11 \leq 10$ (False)

X

X

print the odd number from 1 to 5

```
for (i=1; i<=5, i++) {
```

```
    if (i%2!=0) {
```

```
        print(i)
```

```
    }
```

```
}
```

Initialization

condition

logic

update

i = 1

1 ≤ 5 (true)

if (1%2!=0 (T))
print(1)

i = i + 1
= 1 + 1
= 2

i = 2

2 ≤ 5 (true)

if (2%2!=0 (F)) = 2 + 1 = 3

i = 3

3 ≤ 5 (true)

if (3%2!=0 (T)) = 3 + 1 = 4
print(3)

i = 4

4 ≤ 5 (true)

if (4%2!=0 (F)) = 4 + 1 = 5
No

i = 5

5 ≤ 5 (true)

if (5%2!=0 (T)) = 5 + 1 = 6
print(5)

i = 6

6 ≤ 5 (false) x

print the reverse number 5 to 1

5 4 3 2 1

```
for (i=5, i >= 1, i--){
```

```
    print(i)
```

```
}
```

Initialization

condition

logic

update

i = 5

5 >= 1 (T)

print(5)

i = 5 - 1 = 4

i = 4

4 >= 1 (T)

print(4)

i = 4 - 1 = 3

i = 3

3 >= 1 (T)

print(3)

i = 3 - 1 = 2

i = 2

2 >= 1 (T)

print(2)

i = 2 - 1 = 1

i = 1

1 >= 1 (T)

print(1)

i = 1 - 1 = 0

i = 0

0 >= 1 (F)

~~print(0)~~

X

factorial of number using for loop:

let $n = 3$

let $mul = 1$

for (let $i = 1$, $i \leq n$, $i++$) {

$mul = mul * i$

}

console.log(mul)

Initialization

Condition

logic

update

$i = 1$

$i \leq 3$

$mul = mul * i$

$i = i + 1$

$i = 1$

$1 \leq 3$

$mul = 1 \times 1$
 $= 1$

$i = 1 + 1 = 2$

$i = 2$

$2 \leq 3$

$mul = 2 \times 1$
 $= 2$

$i = 2 + 1 = 3$

$i = 3$

$3 \leq 3$

$mul = 3 \times 2$
 $= 6$

$i = 3 + 1 = 4$

$i = 4$

$4 \leq 3$ (F)

~~mul~~

~~i~~